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DAV Assignmnet_02

INTRODUCTION:

Purpose of Assignment:

The purpose of this assignment is to design and implement two interactive data-driven dashboards using D3.js, demonstrating the ability to work with spatial, temporal, network, and hierarchical visualizations.

The dashboards aim to convert large real-world datasets into meaningful insights through interactivity, animation, and linked views. The assignment evaluates skills in data preprocessing, visualization design, modular architecture, and user interaction.

Goals & High-Level Learning Outcomes:

By completing the assignment, the following learning outcomes are achieved:

- Understand and implement core D3.js modules, including scales, shapes, forces, brushing, transitions, and geographic projections.
- Developing interactive dashboards using multiple coordinated views.
- Apply data wrangling, cleaning, grouping, and parsing to large datasets.
- Create meaningful analytical tools such as zoomable map, force-directed clustering, brushable timeline, motion chart, choropleth map, and sunburst hierarchy.
- Ability to communicate insights visually through animation, filtering, and tooltips.

Overview of the 2 dashboards:

This project consists of two separate dashboards, each designed to highlight a different aspect of global datasets:

Task 1 – Spatial & Network Analysis Dashboard:

A dashboard visualizes global power plant locations using map-based geographic visualizations, force layouts, and a timeline.

Features:

- Geographic aggregation with semantic zoom
- Force-directed clustering by fuel type
- A linked brushable timeline

Task 2 – Temporal & Hierarchical Simulation Dashboard:

A dashboard based on Gapminder, showing the evolution of countries over time.

Features:

- Motion chart of GDP vs Life Expectancy
- Animated play/pause timeline
- Choropleth map
- Sunburst hierarchical population breakdown

DATA DESCRIPTION:

Global Power Plant Database (Task 1)

Source: Global Power Plant Database (World Resources Institute)

Columns Used:

- latitude
- longitude
- fuel
- capacity_mw
- country
- commissioning_year

Data Size: 35,000 rows after cleaning.

This dataset is ideal for spatial and cluster-based visualizations because it contains both geographic attributes and categorical attributes (fuel type).

Gapminder Dataset (Task 2)

Columns Used:

- country
- continent
- gdp
- lifeExp
- pop
- year

This dataset supports time-based analyses and multivariate visualization through scatter plots, choropleths, and hierarchical breakdowns.

Preprocessing Steps:

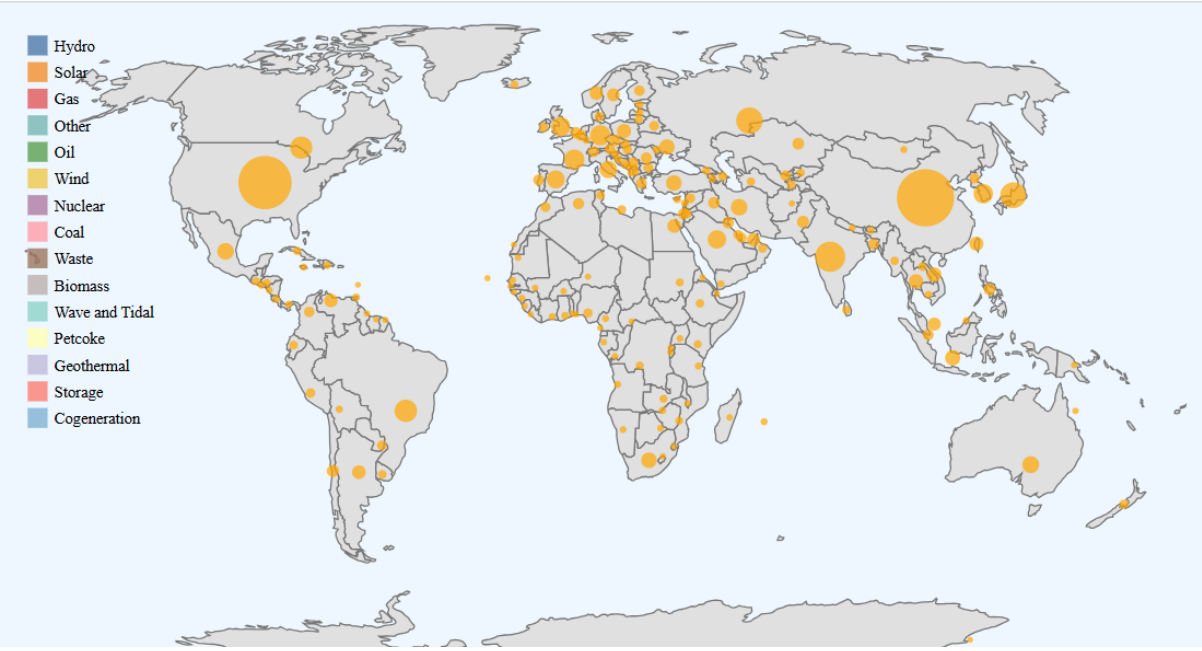
The Global Power Plant Database required no external preprocessing, as all filtering, type conversion, and aggregation were handled directly in D3.js during visualization. Only the Gapminder dataset required basic preprocessing in Python to convert wide tables into long format, merge indicators, correct country names, and attach continent information. Using Python (Pandas), the population, GDP per capita, and life expectancy files were loaded and

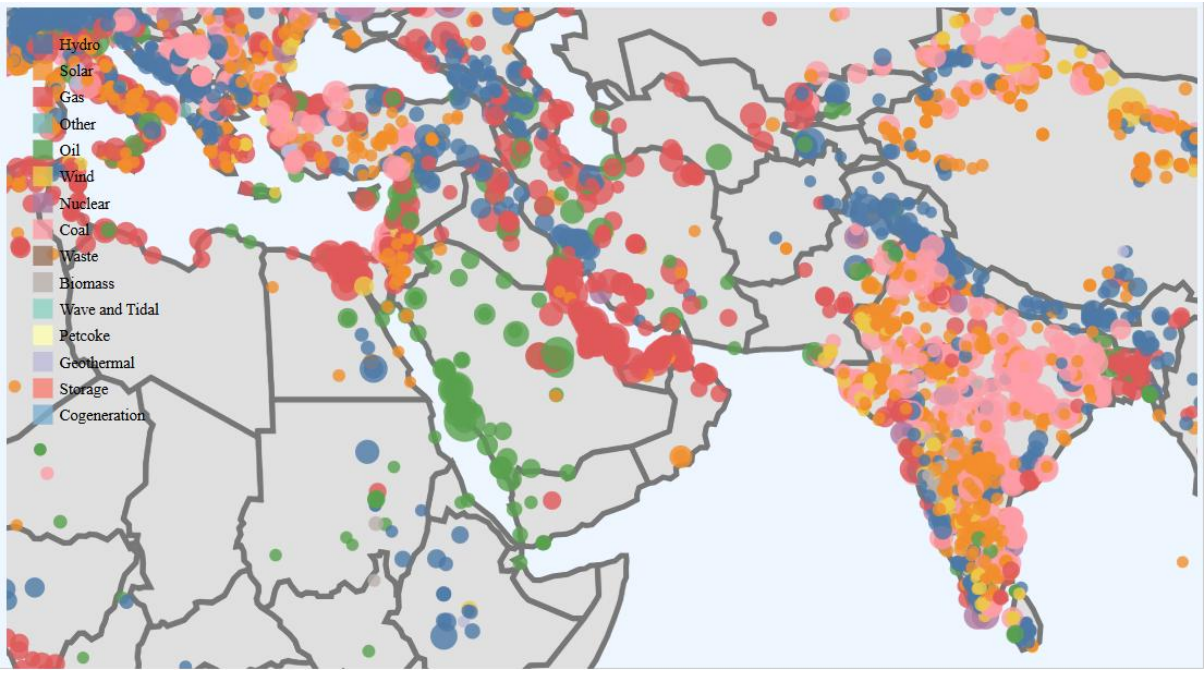
converted from wide to long format using melt, after which the year column was converted to an integer, and all three datasets were merged on geo, name, and year. A small set of country name corrections was applied to ensure a successful merge with the continent mapping file, producing a clean final dataset saved as gapminder_combined.csv. The Global Power Plant Database used in Task 1 required no external preprocessing; only basic type conversions and removal of rows with missing coordinates were performed inside the D3 code during visualization. All remaining transformations—such as grouping by country, filtering by fuel, aggregating capacities, and applying time-based filters—were computed dynamically inside D3 during rendering rather than handled as preprocessing.

Task 1 — Spatial & Network Analysis Dashboard:

Requirement 1.1 — Zoomable Aggregate Map

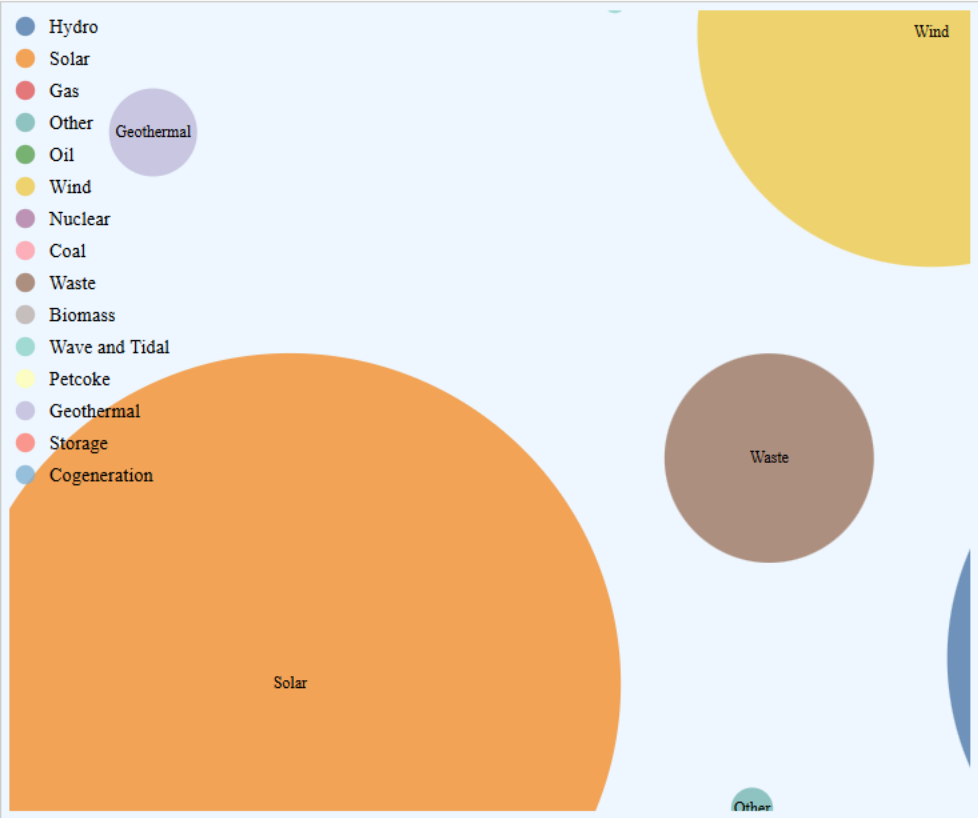
The zoomable aggregate map was built using Mercator projection combined with D3's zoom behavior to allow users to explore global power plant distribution at different scales. When zoomed out, the visualization displays aggregated circles at the country level, where bubble size represents the total installed electricity capacity, enabling a high-level overview without overwhelming the user. As the user zooms past a predefined threshold, semantic zooming replaces country bubbles with individual plant markers rendered at their actual coordinates, each coloured by fuel type and equipped with detailed tooltips. To maintain smooth interaction despite the dataset's huge rows, several optimizations were implemented, including caching aggregated values, removing outdated layers before re-rendering, minimizing redraw costs through SVG layering, and smoothing zoom transitions.



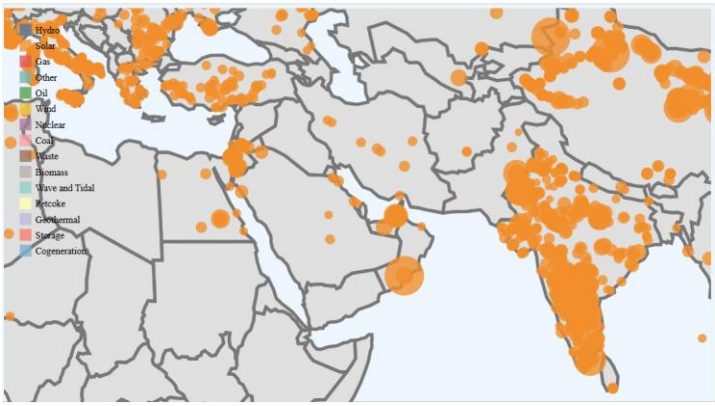


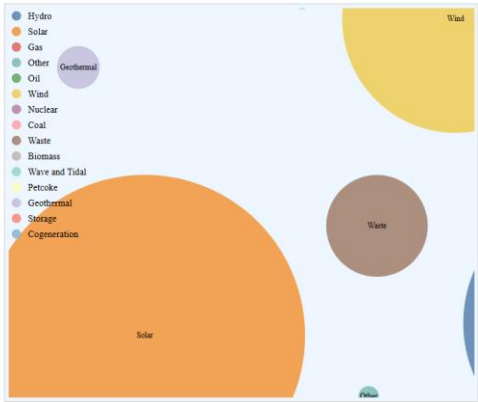
Requirement 1.2 — Force-Directed Fuel Clustering

The Force-Directed Fuel Clustering visualization presents each fuel type as a bubble whose size represents the number of power plants using that fuel, arranged via a physics-based force simulation. A combination of forces, forceCollide for spacing, forceCenter for layout stability, and directional forceX/ forceY for grouping, ensures clear separation between clusters. Each fuel is assigned a distinct color using a large categorical palette to avoid repetition among the many fuel categories, such as Hydro, Solar, Wind, Nuclear, and others. The chart acts as an interactive global filter; clicking a fuel node immediately updates the map and timeline to display only the selected fuel type, ensuring full synchronization across views. This interactivity required handling name inconsistencies and preventing node overlap through collision forces.



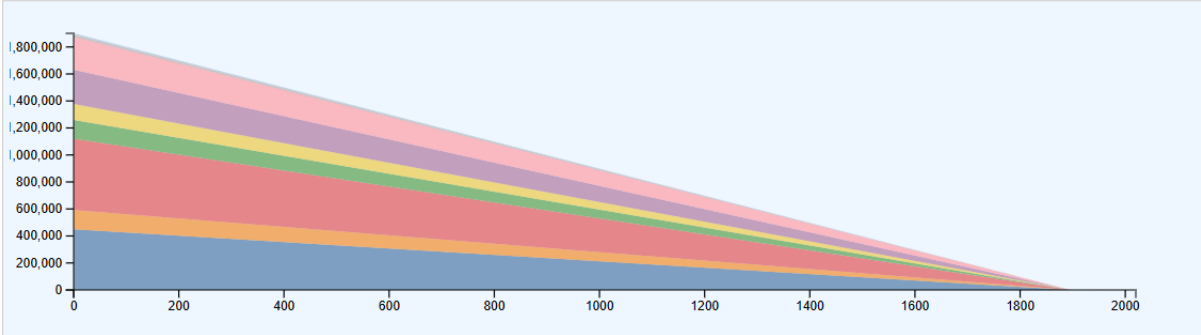
Clicking solar impact on the map

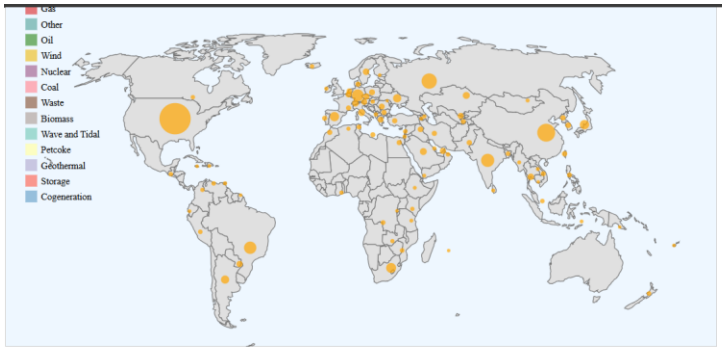


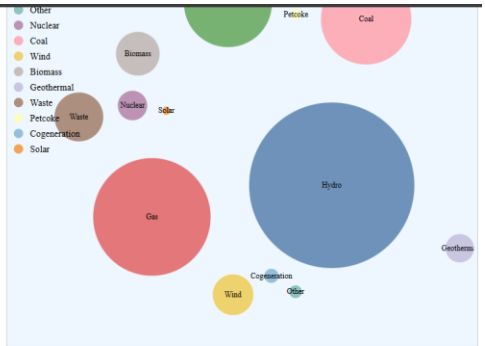


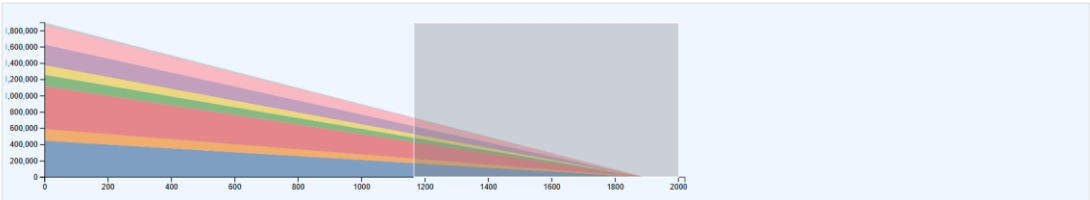
Requirement 1.3 — Brushable Timeline

The Brushable Timeline is a stacked area chart representing the number of power plants commissioned per year, built to allow users to filter the entire dashboard by selecting a year range using a D3 brush. When the user drags across the timeline, the brush returns a [minYear, maxYear] range, which dynamically updates both the global map and force-directed clustering in real time. Only plants commissioned in the selected period remain visible across all views. Smooth brushing, cached filtering, and a custom-designed brush handle were implemented to ensure responsive performance during interaction. This timeline acts as the temporal backbone of Task 1, providing intuitive control over the dataset's historical distribution.





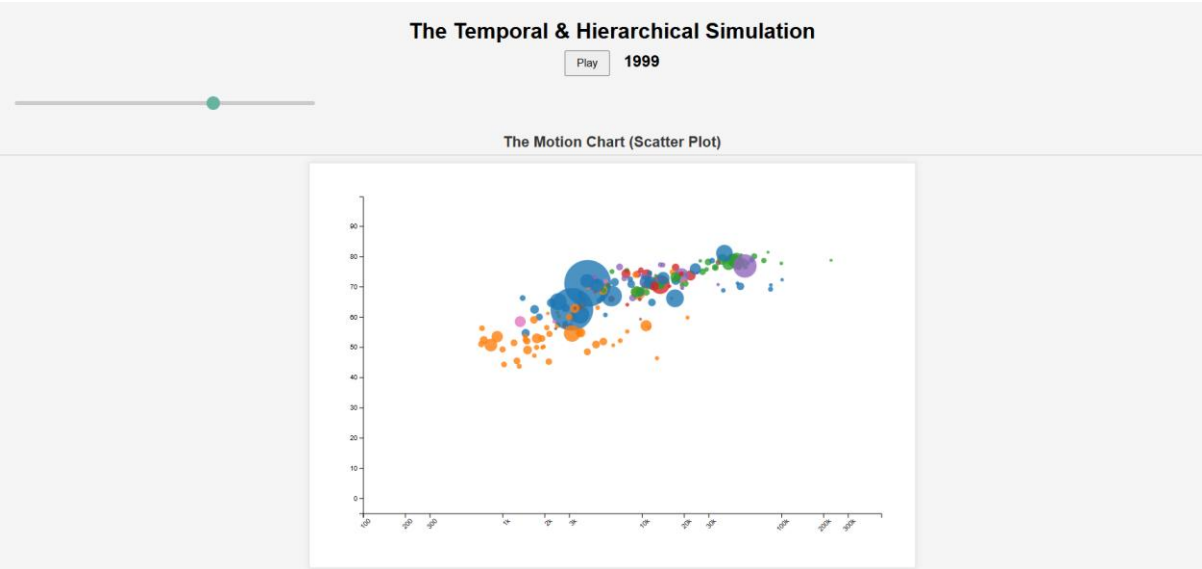




Task 2 – Temporal & Hierarchical Simulation Dashboard:

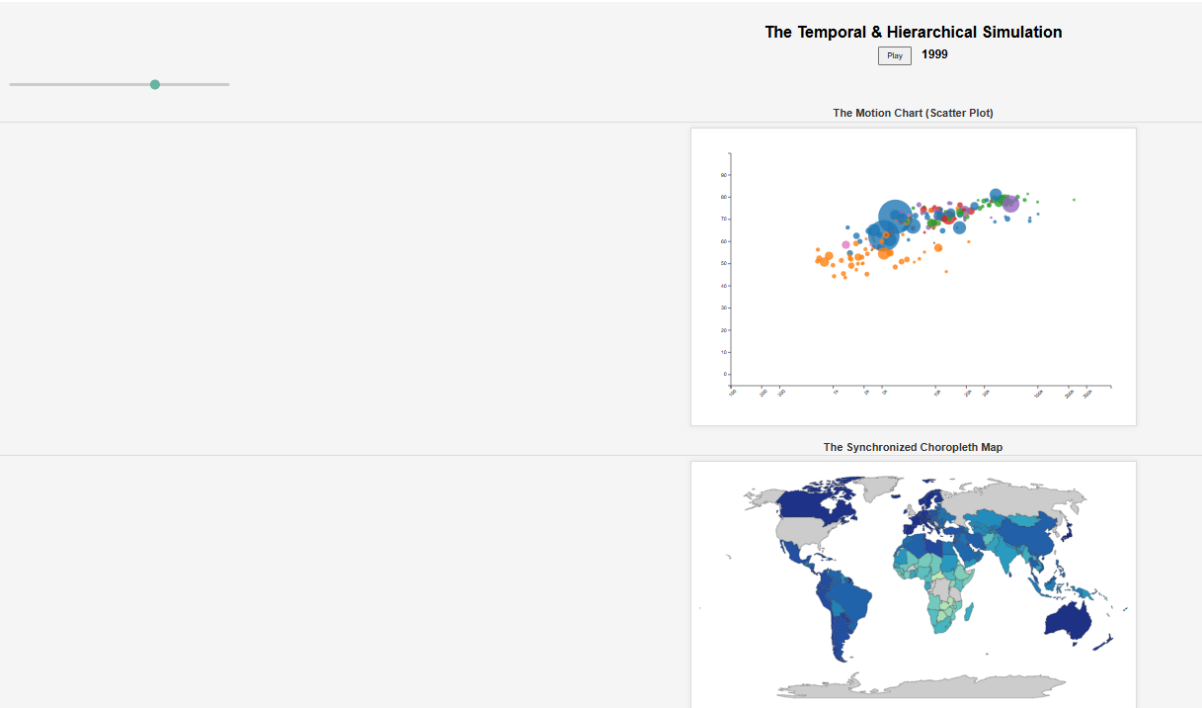
Requirement 2.1 — Temporal Motion Chart (Scatter Plot)

The Motion Chart visualizes the relationship between GDP per capita and life expectancy for countries over the last 2 decades. Each bubble represents a country, with its position determined by GDP (logarithmic X-axis) and life expectancy (linear Y-axis), and its size proportional to population. Bubbles are colored by continent, allowing users to quickly identify regional patterns. The chart supports animation via a Play/Pause button, automatically advancing through years with smooth transitions, highlighting historical changes in wealth and health. The dynamic representation is complemented by a draggable slider, enabling the handling of hundreds of countries by updating only relevant elements per year and caching scales to maintain smooth performance.



Requirement 2.2 — Synchronized Choropleth Map

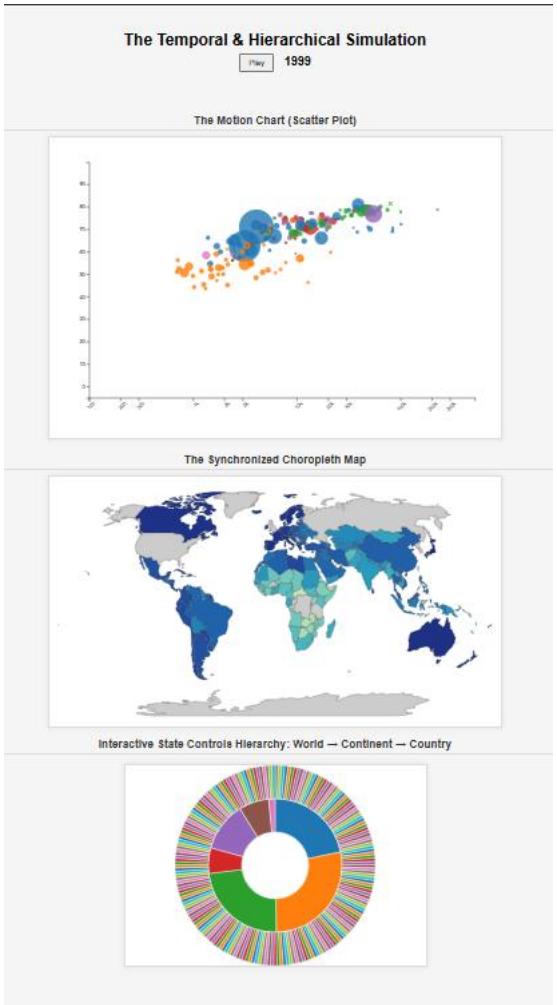
The Choropleth Map is strictly synchronized with the Motion Chart, coloring countries according to life expectancy for the currently selected year. A natural Earth projection with d3.geoPath ensures accurate global representation, while a sequential color scale visually distinguishes life expectancy ranges. As the Motion Chart progresses through years, the map updates in real time, reinforcing temporal insights. Hover interactions allow users to inspect specific countries, providing precise life expectancy values. This coupling enables users to observe regional trends in health outcomes in concert with economic indicators, enhancing the dashboard's analytical depth.



Requirement 2.3 — Interactive State Controls

Interactive state controls enhance the temporal and hierarchical exploration of the dataset. The draggable year slider allows users to scrub through years manually, immediately updating the scatterplot and choropleth map while pausing the ongoing animation. The hierarchy chart, implemented as a Donut or Sunburst, presents the data at three levels: World --> Continent --> Country.

Hovering over a continent highlights only the relevant countries in the scatter plot, reducing visual clutter and emphasizing regional patterns. These controls provide users with both overview and detail-on-demand capabilities, supporting flexible analysis of temporal and geographic trends across more than a century of data.



INTERACTIONS & LINKED VIEWS:

All components are interconnected to create a fully coordinated dashboard experience. In Task 1, selecting a fuel type in the Force-Directed Clustering immediately filters the Map and Timeline to display power plants of that fuel. Similarly, brushing a year range on the Timeline dynamically updates both the Map and Force Layout, showing only plants commissioned within the selected period. In Task 2, changing the year via the Motion Chart animation or

the draggable slider simultaneously updates the Scatter Plot, Choropleth Map, and Sunburst Hierarchy. Hover interactions on the hierarchy emphasize only relevant countries in the Scatter Plot, providing both global context and drill-down insights.

API USED:

D3-geo: used to create geographic visualizations in both dasboards. In Task 1, it handled the world map projection and rendering of country-level and individual powerplant points. In Task 2, it was used to render the choropleth map showing life expectancy per country.

D3-zoom: Applied in Task 1 to allow users to zoom in and out on the map. Sementic zoom logic used this module to switch between country aggregates and individual power plants as users zoomed.

D3-drag: Used in Task 2 to create the draggable year slider. Users culd scrub through time manually, pausing or oving the animation to explore specific years.

D3-brush: Implemented in Task 1 for thew brushable timelne. Users can selecta range of years, and the selection dynamically updates thre map and force-directed fuel clusters to show only power plants commissione in that period.

D3-fetch: Used to load the Global Power Plant CSV data asynchronously.

D3-force & D3-scale: Applied to create the Force-Directed Fuel Clustering. forceCollide prevented node overlap, forceCenter centered the layout, and forceX/forceY grouped nodes by fuel type. d3.scaleOrdinal assigned unique colors to each fuel type.

D3-shape & D3-axis: The stacked area chart was created with d3.area (shape), the X/Y axes were rendered using d3.axisBottom/d3.axisLeft, and d3.brushX enabled interactive year range selection that updated the map and force layout.

D3-timer, D3-transition, D3-interpolate: Implemented the Motion Chart animation. These APIs allowed smooth transitions of bubbles over time, interpolating positions and sizes as the year advanced automatically or via the slider.

D3-hierarchy: d3.hierarchy supported the sunburst hierarchy chart for World → Continent → Country drill-down interactions.

CONCLUSION:

In the assignment, two interactive dashboards were successfully developed using D3.js, demonstrating the integration of spatial, temporal, network, and hierarchical visualizations on real-world datasets. Task 1 showcased global power plant distributions through a zoomable map, force-directed clustering by fuel type, and a brushable timeline, enabling users to explore both geographic and temporal patterns interactively. Task 2 highlighted the evolution of countries' wealth and health via a motion chart, synchronized choropleth map, and hierarchical sunburst, providing insights into economic and demographic trends over time. Throughout the project, D3.js APIs including d3-geo, d3-zoom, d3-drag, d3-brush, d3-force, d3-scale, d3-timer, and d3-hierarchy were leveraged to create smooth, coordinated,

and responsive visualizations. The dashboards not only facilitated data exploration and comparison across multiple dimensions but also demonstrated best practices in performance optimization, interactivity, and visual storytelling, making complex datasets accessible and insightful for real-world analysis.

REFERENCES:

- **Global Power Plant Database (WRI):** https://github.com/wri/global-power-plant-database
- **Gapminder Data:** https://www.gapminder.org/data/